IS2209 Data Governance and Management Course

# Data Governance & Management Framework

## Michelin Sri Lanka Manufacturing Plant Kotugoda

Presented by

**Group 14**

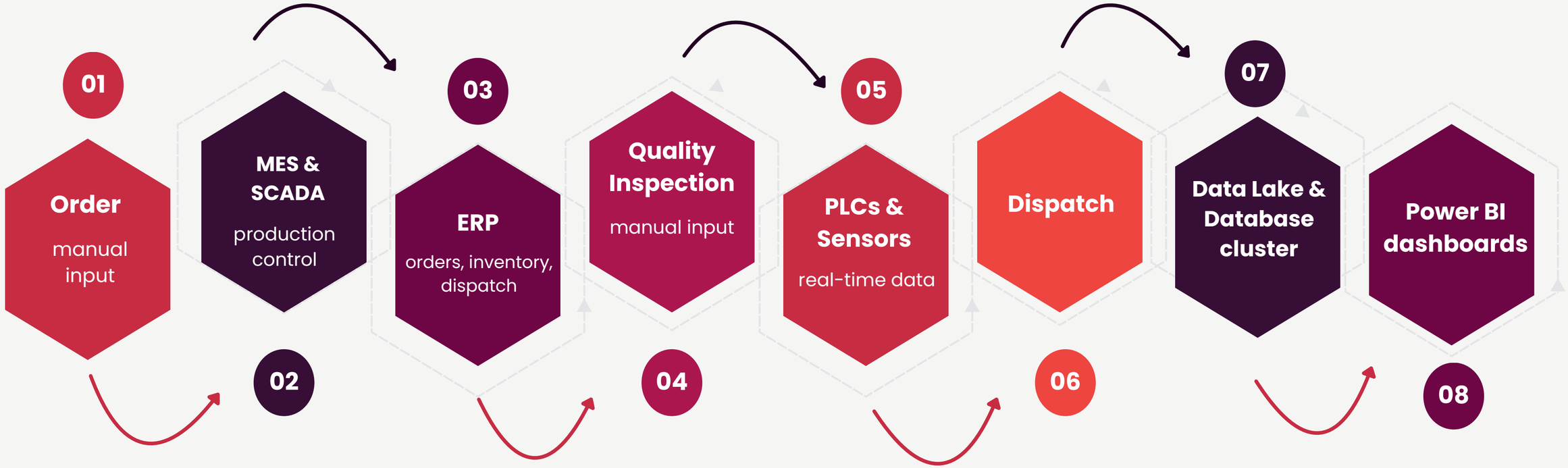
# Organization Overview

- **THE CONTEXT**
  - Specialized manufacturing plant producing 800+ tyre variations (solid tyres, tracks, and industrial wheels).

- **OPERATIONAL MODEL**
  - Data-intensive
  - make-to-order production requiring strict quality control and traceability.

- **CURRENT PRACTICES**
  - Basic system-specific controls
  - Role-based access for recipes
  - Serial-number traceability already exists.

# End-to-End Dataflow



# Key Data Management Challenges

- **Lack of Standardization** - Non-uniform naming conventions across machines and factories.
- **Structural Inconsistency** - Differing data structures and acronyms for similar parameters.
- **Integration Hurdles** - Challenges in connecting legacy systems with modern analytics.
- **Dependency Risks** - High reliance on complex data dictionaries and limited local support personnel.
- **Security Needs** - Increasing requirements for cybersecurity investment to protect intellectual property.

# Why Governance?

**WITHOUT GOVERNANCE**

- Data inconsistency across systems
- Quality & production risks
- Intellectual Property exposure
- Inefficient decision-making

**WITH GOVERNANCE:**

- Standardization
- Security
- Trusted analytics

# Framework Design

# Goals & Focus Domains

- Primary Goal - Create a secure, scalable approach to manage data, ensuring high quality and protection of intellectual property.

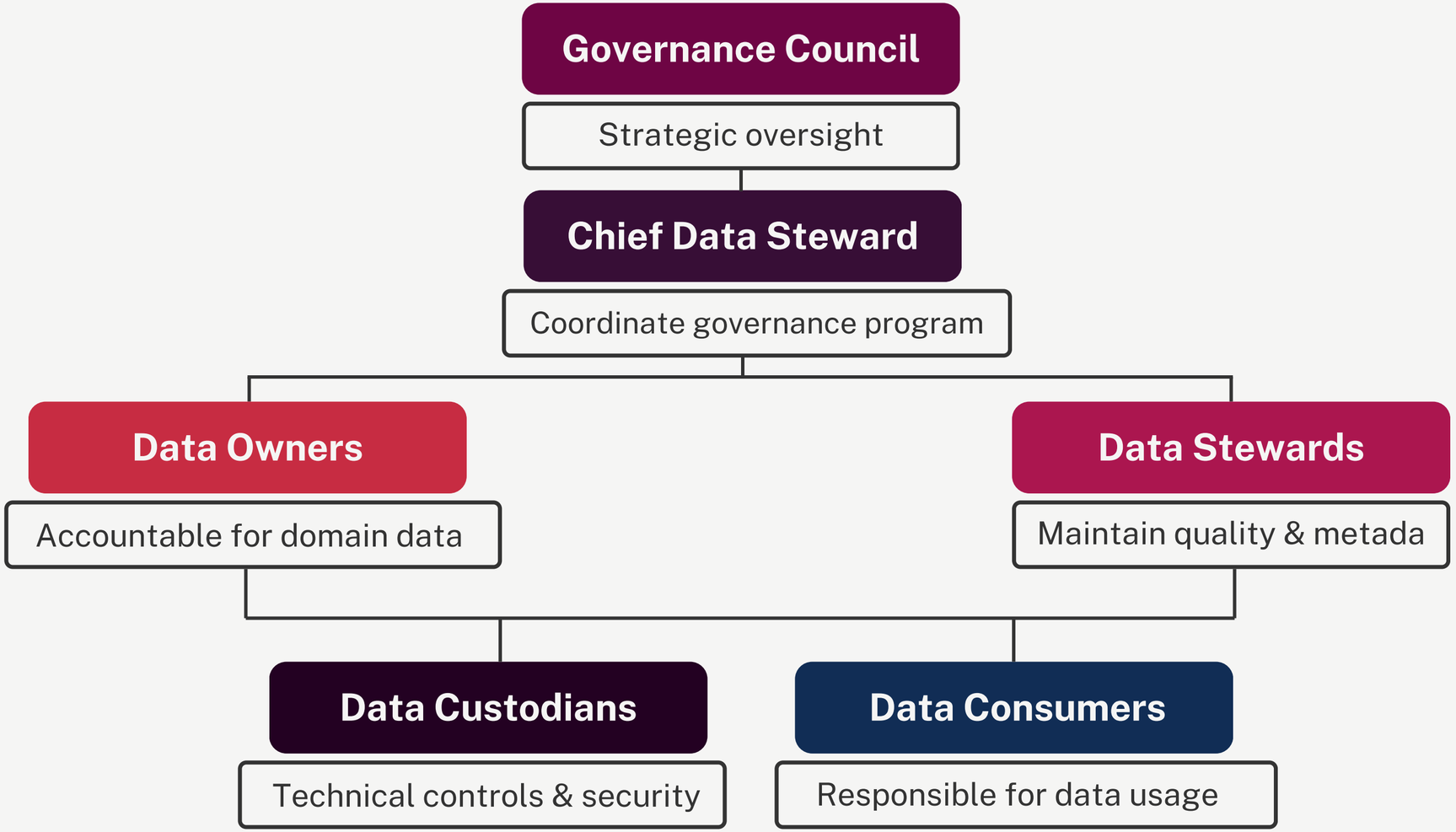
- Key Focus Domains:

  - **Data Security** - Protecting sensitive recipe data.

  - **Data Quality** - Ensuring reliable manufacturing analytics.

  - **Data Architecture** - Unifying interconnected IT and OT systems.

# Roles & Accountability



# Key Policies & Standards

- **Data Classification policy:**
  - **Super Critical**: Recipes/Formulations (Permanent retention).
  - **Critical**: Production data/Quality results (7-10 years retention).
  - **Normal**: Aggregated KPIs/Reports (3-5 years retention).

- **SECURITY POLICY:**
  - Principle of least privilege
  - Role-Based Access Control (RBAC)
  - Dual authorization for Super Critical data.

- **NAMING CONVENTIONS**
  - Uniform codes for equipment, location, units (e.g., "ML_01" for machines).

- **QUALITY STANDARDS**:
  - Real-time production data must be available within 30 seconds
  - Standards for accuracy and completeness

# Architecture & Quality

- **Layered Architecture**
  - Seamless flow from Sensors →PLC → SCADA → MES → ERP → Data Lake.
- **Segregation**
  - Maintaining raw, processed, and curated data layers.
- **Master Data Management (MDM)**
  - Establishing "Single Source of Truth" for Product, Recipe, Equipment, and Supplier masters.
- **Quality Controls**
  - Automated dashboards for monitoring completeness (serial numbers mandatory) and consistency across machines.

# Implimentation Plan

**Phase 1**

**0–6 Months**

**Foundation**
Establish Committee, Assign owners/stewards and classify data.

**Phase 2**

**6–12 Months**

**Quality & MDM**
Implement MDM, standardized schemas and automated validation rules.

**Phase 3**

**12–18 Months**

**Analytics Enablement**
Create analytics ready datasets and enable governed self service Power BI.

**Phase 4**

**18+ Months**

**Optimization**
Regular audits, onboarding new factories and predictive maintenance use cases.

# Business Benefits

Higher Product Quality & Traceability

Improved Operational Efficiency

Seamless Data Interoperability Across Systems

Strong Intellectual Property Protection

# Field Visit

# Thank You